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EFFECT OF MATERNAL SEPARATION ON THE COMPLEMENT SYSTEM AND ANTIBODY RESPONSES IN INFANT PRIMATES

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Hemolytic complement activity and complement protein levels were evaluated before and after psychological disturbance in the squirrel monkey. Significant increases in hemolytic complement activity were observed following separation of 6-month-old infants from their mothers. Complement protein levels were generally correlated with hemolytic activity, but did not show the same pattern of change. C3 levels did not change consistently after maternal separation, while C4 decreased significantly on Day 7, and then began to return toward basal levels by Day 14. The alterations in complement activity were associated with elevated adrenal hormone secretion, but were not directly correlated with plasma cortisol levels. A second experiment showed that increased hemolytic complement activity also occurred in juvenile squirrel monkeys following 4-hour to 7-day removals from a peer group. The effect of maternal separation on antibody responses to viral challenge was evaluated in a third experiment. Separated infant squirrel monkeys mounted a smaller antibody response than did control infants that remained with their mothers. A similar decline in antibody responses was observed in separated rhesus macaque infants, but the effect was less marked in this species. Thus, the strong effect of psychological disturbance on immunity in the squirrel monkey is probably related to its prolonged endocrine responses.

Keywords: complement, antibody response, bacteriophage, stress, cortisol, squirrel monkey, maternal separation

A wide variety of stressful events has been shown to influence immune responses in animals and humans (Ader, 1981; Locke & Hornig-Rohan, 1983). Many of these alterations in immunity have been attributed to the immunosuppressive effects of adrenal hormones, although it has been possible to link corticosteroids directly to the immunological measure of interest in only a few studies of psychologically induced stress (Monjan & Collector, 1977; Pavlidas & Chirigos, 1980; Riley, Fitzmaurice & Spackman, 1981). Recent studies have increasingly implicated other hormones, especially the catecholamines and opiate hormones, and the autonomic

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nervous system (Landmann, Muller, Perini, Wesp, Erne & Buhler, 1984; Lloyd, 1984; Shavit, Lewis, Terman, Gale & Liebeskind, 1984). These studies have also delineated selective changes in immune responses that indicate the need to move beyond the concept of a nonspecific immune suppression (Munck, Guyre & Holbrook, 1984). The following work focused on the complement system, one aspect of the immune system that has been largely overlooked in research on psychological factors and immunity. In addition, we utilized the antibody response to viral challenge to provide a second, functional measure of humoral immunity.

Studies on humans have also become increasingly sophisticated in the types of psychosocial variables that are investigated. Changes in immune responses have been correlated with bereavement, loneliness, lack of social support, an inability to express emotion, and mental illness (Levy, Herberman, Lippman & d'Angelo, 1987; Locke & Gorman, in press; Temoshok & Fox, 1984). For example, lympocyte proliferation responses have been shown to be decreased for 1–2 months following loss of a spouse (Bartrop, Luckhurst, Lazarus, Kiloh & Penny, 1977; Schleifer, Keller, Camerino, Thornton & Stein, 1983), and a greater sense of lonelinesss has been correlated with lower natural killer cell activity in medical students and psychiatric patients (Kiecolt-Glaser, Garner, Speicher, Penn, Holliday & Glaser, 1984; Kiecolt-Glaser, Ricker, George, Messick, Speicher, Garner & Glaser, 1984). Given these findings in humans, it is fortuitous that one of the best-studied animal models of psychogenic stress is the response to loss in nonhuman primates. For several decades, researchers have been investigating the behavioral and hormonal sequelae of maternal separation and loss in infant primates (Bowlby, 1973; Mineka & Suomi, 1978). These studies have demonstrated that separated infant monkeys show marked changes in heart rate, body temperature, sleep patterns, cortisol levels, and brain monoamine activity during the first several days after removal from the mother or social group (Coe, Wiener, Rosenberg & Levine, 1985; Kraemer, 1985; Reite & Capitanio, 1985). Studies of immune function in infant macaques following separation indicated further that there is also a decrease in lymphocyte proliferation responses (Laudenslager, Reite & Harbeck, 1982). In addition, a preliminary study by our laboratory showed that maternal separation causes a decrease in circulating levels of complement proteins and antibodies in squirrel monkey infants (Coe et al., 1985).

The current work extended these observations in a more detailed evaluation of humoral immunity. The first two experiment examined changes in the complement system. Complement proteins are important in the defense against bacterial infection and play an important role in inflammatory processes, which may be protective or injurious to the host (Colten, 1980). The complement system consists of a set of proteins that can act sequentially and, to provide a measure of circulating levels, clinical laboratories usually assess two of these proteins—C3 and C4. The functional integrity of the system is usually evaluated by measuring the capacity of serum complement to lyse target cells, i.e., total hemolytic activity. Both complement protein levels and hemolytic activity can be affected by adrenal corticoids (Atkinson & Frank, 1973), and were evaluated before and after separation in our studies on the squirrel monkey.

The antibody response to viral challenge was also determined in control and separated infants to assess another aspect of humoral immunity. Squirrel monkey infants were injected with the bacteriophage ØX174, a benign virus that has been used extensively to evaluate immunodeficiency in humans because it does not cause any clinical symptoms (Peacock, Jones & Gough, 1973; Wedgewood, Ochs & Davis, 1975). Primary and secondary antibody responses were assessed over a 3-

month period. Since it is known that the squirrel monkey shows an unusually high and prolonged adrenal response to separation, we also conducted a comparative study using similar manipulations on the rhesus macaque. Unlike the squirrel monkey infant, which shows elevated cortisol levels for only 2–3 weeks after separation, the rhesus infant usually has increased cortisol levels for only 1–2 days after removal from the mother (Gunnar, Gonazalez, Goodlin & Levine, 1981). Thus, the comparison permitted us to evaluate the possible role of endocrine activation in mediating the immune changes.

METHODS

Subjects and Housing

For Studies I and II on complement, the subjects were 24 mother–infant pairs and 18 juvenile squirrel monkeys (*Saimiri sciureus*) maintained at the Stanford Primate Facility. An additional 13 mother–infant pairs were utilized in Study 3 to assess antibody responses. The infants and juveniles were bred at the laboratory in small maternal groups consisting of 3–4 adult females and their offspring. At 6–7 months of age, infants were removed from their mothers, either for experimental studies or for colony management, and then housed in large, juvenile peer groups. All subjects had health records dating back to birth, with no previous invasive studies. Food and water were available ad libitum; fresh fruit and vitamin supplements were each provided twice per week. The monkeys were exposed to natural lighting and climatic conditions of northern California through windows behind their cages in order to orient their mating season and to provide environmental enrichment.

The 15 rhesus monkey (*Macaca mulatta*) infants evaluated in Study 3 were reared in a large indoor–outdoor cage. To facilitate the rapid collection of blood samples, the mother–infant dyads were transferred into standard indoor cages when infants reached 6–7 months of age. Two mother–infant pairs were housed in adjacent cages, and the maternal separation was imposed by simultaneously removing the mothers after the infants had been acclimated to the cages for 1 week.

Assessment of Complement

For the study of complement, 1.0 ml blood samples were rapidly collected following brief sedation of the squirrel monkeys with ether anesthesia (< 1.0 min). The sample was temporarily kept on ice, and then spun at 2000 rpm for 20 min in a refrigerated centrifuge (4°C). Sera were stored at -70°C until analysis. Total hemolytic complement activity was determined using a standard chromium release assay that was modified for squirrel monkey sera (Rosenberg, Coe & Levine, 1982). Chromated (Cr^{51}) sheep erthrocytes were sensitized with rabbit antisheep erythrocyte antibody and exposed to 10 different dilutions of each serum sample in a total volume of 0.2 ml. After 1-h incubation at 37°C, the ratio of chromium in the cells and supernatant was determined on a Beckman gamma counter. The volume of sera required to cause 50% lysis of the cells was determined by linear regression and converted to CH50 units per ml for analysis.

C3 and C4 levels were measured by single radial immunodiffusion using Kallestad endoplates (Austin, TX). Five μl of sera were added to each well in the agarose gel containing monospecific antisera, and allowed to equilibrate at room temperature for 48 h. The antisera to human complement cross-reacted extensively

and in a linear manner with homologous proteins in primate sera. Complement protein has been reported as a relative measure based on ring size. Ring size was measured with a Transidyne calibrating viewer and the diameter squared (Zone Area).

Assessment of Plasma Cortisol

Plasma cortisol levels were assessed by radioiimmunoassay using antiserum F3-314 from Endocrine Sciences (Tarzana, CA). Each sample was extracted in duplicate and run in triplicate; final values fell within the linear of the standard curve (for further details, see Coe, Murai, Wiener, Levine & Siiteri, 1986).

STUDY I

Procedure

When 24 infant squirrel monkeys reached 6 months of age, they were removed from their mothers and housed in 2 separation conditions. Fifteen infants were housed alone in single cages for 2 weeks (Alone), and 9 were allowed to remain in their homes cages with familiar infants (Group). The Group separation condition was initiated by removing all the mothers from 3 maternal groups and was designed to evaluate the effect of a more benign separation condition. Blood samples were collected 1 week before the separation (Basal) and on Days 1, 7, and 14 after removal from the mother to assess changes in plasma cortisol and the complement system. The results were evaluated with two-way analyses of variance, using a repeated measures design (Separation Condition × Separation Days).

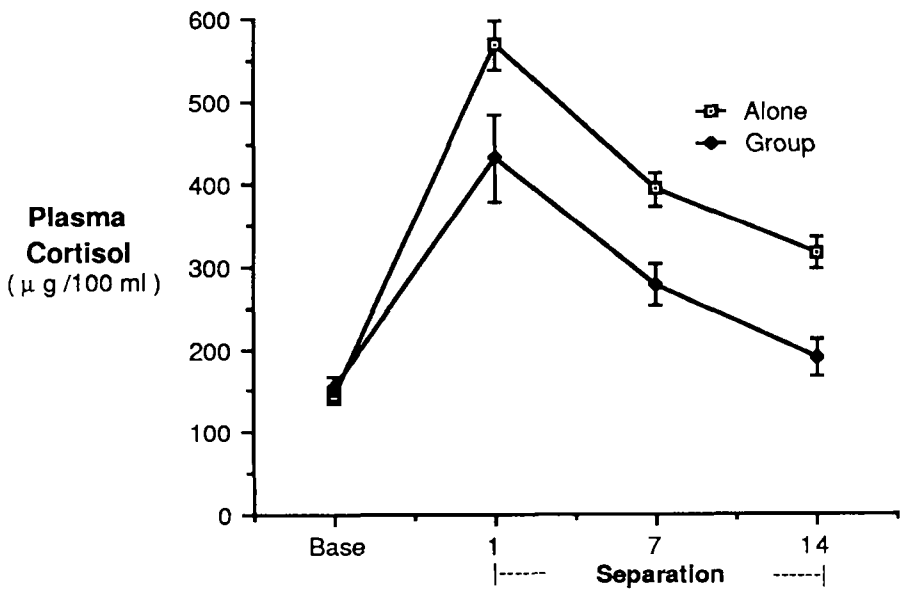


FIGURE 1 Plasma cortisol responses (mean + S.E.) of separated infant squirrel monkeys housed either alone or as a group.

Results

Plasma cortisol levels were significantly elevated in all infants following separation from the mother ($F[3, 66] = 97.11$, $p < .0001$). However, a significant interaction between Time and Separation Condition in the ANOVA indicated that plasma cortisol responses were higher in the infants separated in the Alone condition ($F[3, 66] = 5.24$, $p < .003$) (Figure 1). Alone infants had cortisol levels that were 135 μg/dl over Group infants at 24 h, and they were still 163 μg/dl over Basal levels after 14 days, whereas Group-housed infants had returned to normal levels of adrenal activity.

Infant squirrel monkeys showed the extremely high levels of hemolytic complement activity characteristic of this species (Figure 2). Moreover, the infants showed dramatic increases in hemolytic activity following maternal separation. Many infants showed rapid increases within 24 h, and hemolytic activity was significantly elevated by 7–14 days ($F[3, 66] = 16.26$, $p < .001$). For reasons that could not de determined, 9 of the 15 infants in the Alone condition were slower to show increases and were not elevated on Day 1. However, 13 of the 15 Alone infants and all 9 Group-housed infants showed the markedly elevated hemolytic activity on Day 14.

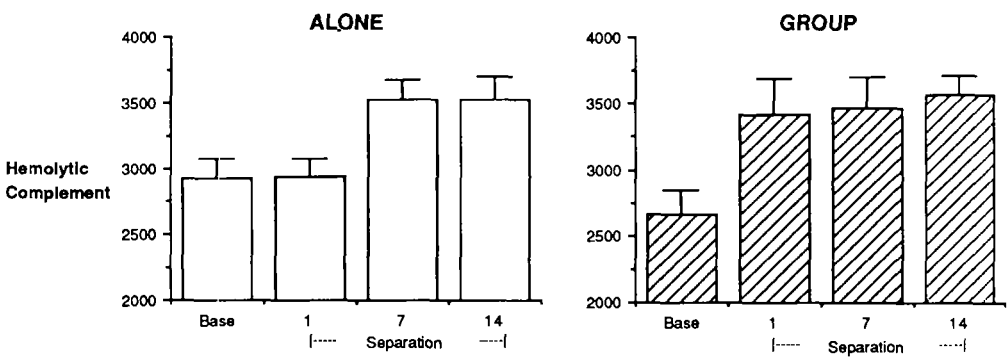


FIGURE 2 Hemolytic complement activity (mean + S.E.) in infant monkeys before and after separation from the mother.

The increased hemolytic activity was not explicable in terms of the changes in complement protein levels. Although C3 and C4 levels were generally correlated with hemolytic activity in baseline samples ($r = .43$ and .40, respectively), neither protein increased in response to maternal separation. C3 levels fluctuated across the separation, but did not change significantly in either separation condition (Figure 3). C4 levels actually decreased significantly after 7 days of separation in both housing conditions and then returned toward basal levels by Day 14 ($F[3, 66] = 2.74$, $p < .01$). Attempts to correlate changes in the complement system with plasma cortisol levels did not reveal any significant relationship beyond the general temporal association of these shifts.

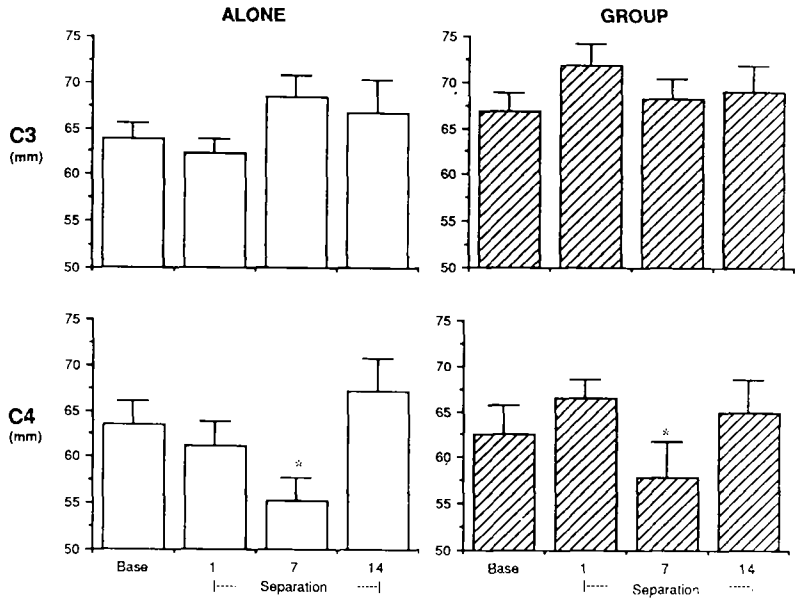


FIGURE 3 Complement protein levels (mean + S.E.) in infant monkeys before and after separation from the mother.

STUDY II

Procedure

Study II evaluated whether juvenile squirrel monkeys would show a similar change in hemolytic activity following removal from a peer group. Six monkeys were assigned to a 4-h removal condition; 6 monkeys were removed from the peer group for 2 days; and 6 monkeys were removed from the group for 7 days. In addition, to verify that the effect was not caused by repeated collection of blood samples, the order of the Base and Separation samples was counterbalanced across subjects. Basal blood samples were collected either 2 weeks before or 2 weeks after the separation. During the separation, 2–4 juveniles were housed individually in separate cages similar to those used in the infant experiments, and then they were returned to the peer group. The results were analyzed in a two way analysis of variance, using a repeated measures design (Three Separation Durations × Base/Separation).

Results

Baseline hemolytic activity was higher in juveniles than in infants, but the effect of social separation was similar (Figure 4). Significant increases in hemolytic activity were observed after removal of juveniles from the peer group, and the effect was rapid and consistent ($F[1, 15] = 6.63$, $p < .02$). By 4 h, 4 of the 6 monkeys showed increases over basal values; by 2 and 7 days, 5 of 6 monkeys in each condition showed marked elevations over basal values. Circulating levels of C3 and C4 did not change significantly in these older juveniles, and the plasma cortisol response to peer separation was of a shorter duration, returning to basal levels by Day 7.

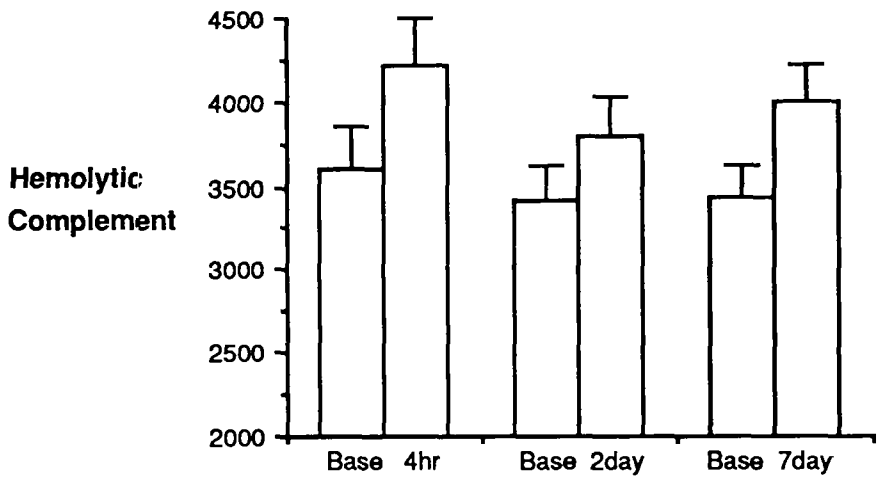


FIGURE 4 Hemolytic complement activity (mean + S.E.) in juvenile monkeys following removal from their social group for 4h to 7 days.

STUDY III

Procedure

The third study evaluated the effects of maternal separation on the antibody response to viral challenge in squirrel monkey and rhesus monkey infants. Thirteen 6-month-old squirrel monkey infants were used to evaluate the antibody responses to ØX174. Six infants served as controls and were injected while they remained undisturbed with their mothers (Control). Four infants were separated from their mothers and housed alone for 7 days before they were injected with the phage (Separated). Three additional infants were separated alone for 7 days, but then reunited with the mother on the day of phage injections (Sep-Reunion). Primary antibody responses were assessed in blood samples collected on Days 7, 14, 28, and 56 after injection of the phage. On Day 56, a booster injection was administered, and secondary antibody responses were assessed after 7 and 14 days to evaluate immunological memory.

A similar experiment was conducted on rhesus macaque infants. When the infants reached 6–7 months of age, four were separated from their mothers for 7 days and then injected with the phage (Separated). The primary and secondary antibody responses were compared to four infants that remained undisturbed with their mothers (Control). In order to be certain that the separated rhesus infant showed a smaller adrenal response to separation than did the squirrel monkey, blood samples were also collected during the week prior to injection of the phage in both control and separated monkeys (at 0900 and 1530 on Days 1, 3, 4, and 7). A replication of the experiment was also run after the assessment of antibody responses revealed a smaller effect of separation in this species. Four more infants were evaluated in the separation condition, and three served as contemporaneous controls.

Assessment of Antibody Responses

The bacteriophage was acquired from American Type Culture Collection (ATCC 13706 B1, Rockville, MD) and propagated on *E. coli.* Phage titer was determined by plaque count, and a concentration of 4.5×10^9 plaque-forming units (pfu) was used for injection. Each monkey was injected with 0.1 ml in the saphenous vein; primary and secondary antibody responses were then evaluated with a neutralization assay (modified from Peacock, Jones & Gough, 1973; Stashak, Baker & Robinson, 1970). Serum samples were serially diluted in nutrient broth (Oxoid CM67) and 90 μl of the six dilutions were incubated with 100 μl of phage (125–200 pfu) for 24 h at 8°C. The phage was then propagated on a lawn of *E. coli* C and the relative level of neutralization determined by comparing the plaque titer in quadruplicate drops to an expected count (25–30 plaques). Neutralization values between 10 and 90% were used to generate a regression line after log conversion of the serum volumes, and the 50% neutralization point was calculated as an index of antibody titer (SD50). For each time point, the median (Q1 and Q3) volume of serum required to reach SD50 is illustrated in the graphs.

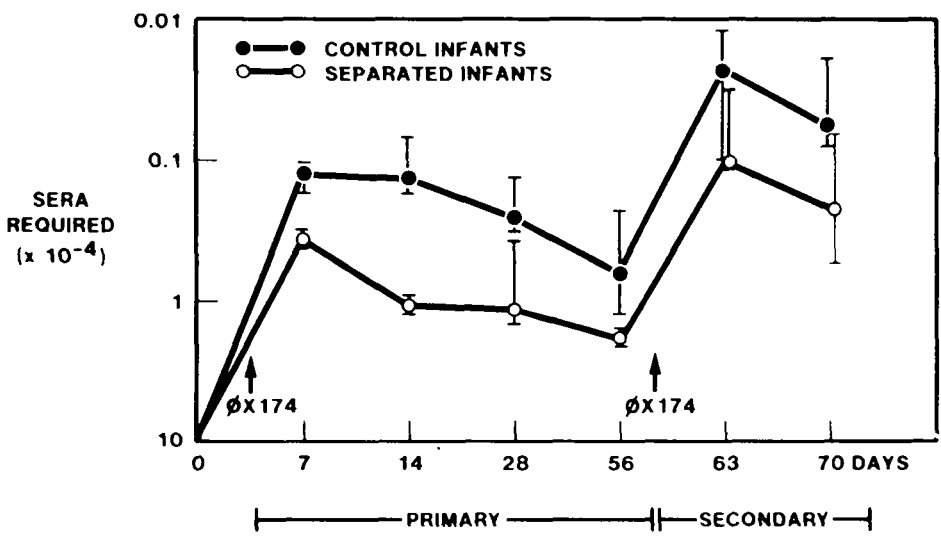


FIGURE 5 Primary and secondary antibody responses (median + Q) of control and separated squirrel monkey infants. Higher antibody levels are indicated by smaller serum volumes (μl) required to neutralize the bacteriophage in the assay.

Results

All squirrel monkeys mounted vigorous antibody responses to the phage, but the primary antibody response was lower in separated infants than in control monkeys ($F[1, 8] = 5.99$, $p < .04$). On Day 14, when primary responses peaked, there was no overlap in the serum neutralizing values from control and separated subjects (Figure 5). The average neutralization values remained disparate on Days 28 and 56, although individual scores began to overlap as all monkeys showed a decline in

antibody titers. Following the booster injection of the phage, the infants showed dramatic increases in neutralizing antibody indicative of secondary responses, and the difference between control and separated subjects was nonsignificant. It was also of interest that the primary antibody responses of the 3 infants tested in the Sep-Reunion condition were more similar to continuously separated subjects than to controls. The small sample size makes this observation preliminary, but on Day 14, when the primary response peaked, sera from the 3 reunited infants showed neutralizing capacity below that of all 6 control infants.

As can be seen in Figure 6, separated rhesus infants showed a similar tendency for lower levels of neutralizing antibody when compared to undisturbed, control infants. The difference in the primary response reached statistical signficance ($F[1, 6] = 7.55$, $p < .03$], but the overlap in individual values led us to replicate the study. Evaluation of 7 additional infants revealed the same tendency for lower antibody titers as in the 4 separated subjects, but again with an overlap in values. The smaller effect of separation in the rhesus infant may have been related to their attenuated adrenal response to separation. Plasma cortisol levels were not elevated during the week of separation and had already returned to normal levels by 24 hours after removal of the mother (Figure 7).

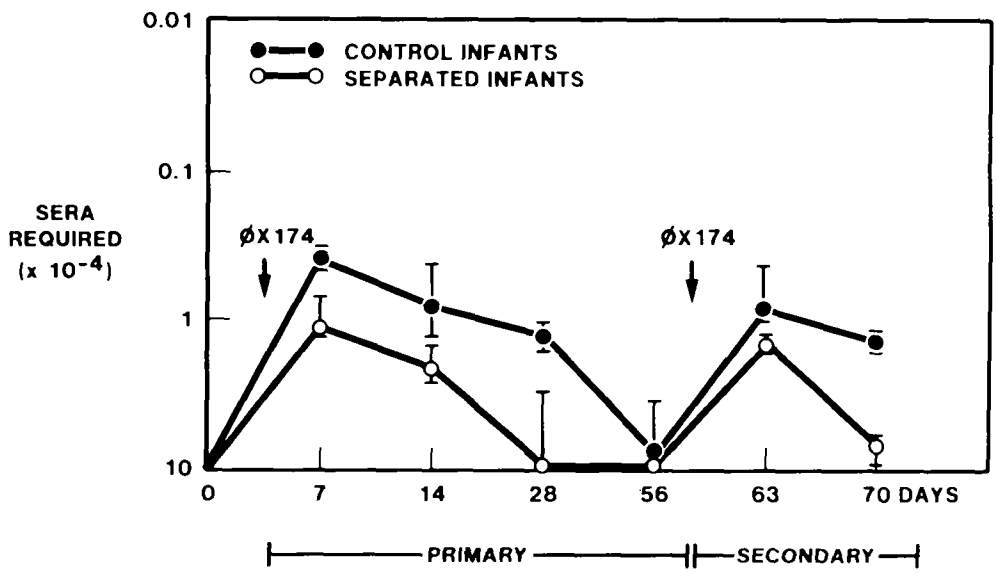


FIGURE 6 Primary and secondary antibody responses (median + Q) of control and separated rhesus monkey infants. Higher antibody levels are indicated by smaller serum volumes (μl) required to neutralize the bacteriophage in the assay.

DISCUSSION

These results show that the psychological disturbance evoked by maternal separation can alter humoral immunity in the infant monkey. The decrease in antibody responses to ØX174 is in keeping with the inhibitory effect of noise-novelty stress on antibody responses to bovine serum albumin in juvenile

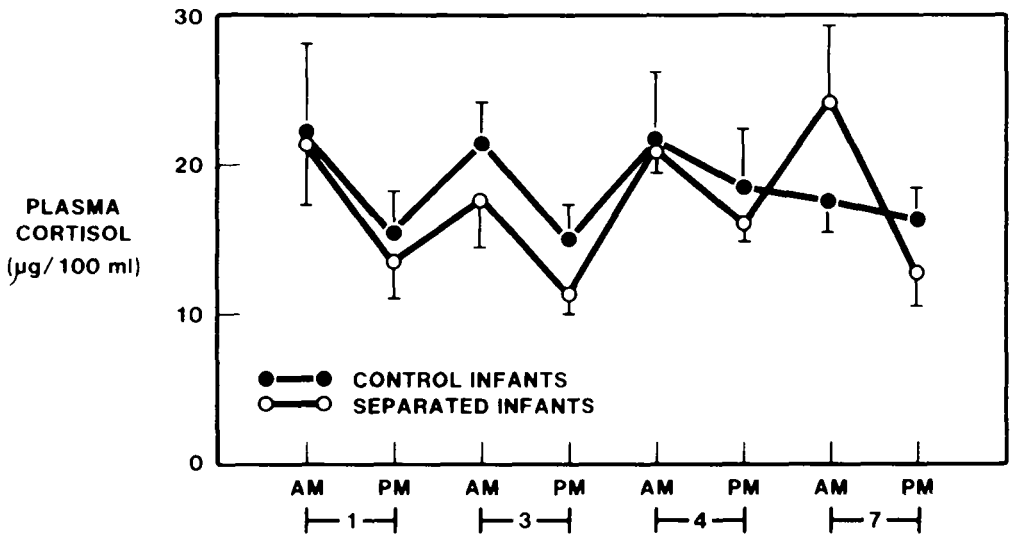


FIGURE 7 Plasma cortisol levels (mean + S.E.) in control and separated rhesus monkey infants, showing that separation did not evoke a sustained increase in adrenal activity in this species.

vervet monkeys (Hill, Greer & Felsenfeld, 1967) and concurs with a prior report describing lower lymhocyte proliferation responses following separation in infant pigtail monkeys (Laudenslauger, Reite & Harbeck, 1982). The complement system has not been studied previously from this perspective, and a sustained increase in hemolytic complement activity may initially appear to be a surprising finding.

Several factors need to be considered in order to understand the changes in the complement system. First, it should be noted that Atkinson and Frank (1973) reported a 20% increase in hemolytic activity and a 51% in C1 titers lasting for 2 weeks when guinea pigs were administered corticoids within the normal physiological range (cortisone acetate: 5 mg/kg/day). Subsequent citations of this study focused only on the inhibitory effects observed after corticoids were administered at high, pharmacological levels (20–100 mg/kg/day). The issue of corticoid level and species is important; 10 mg of hydrocortisone in the mouse caused a decrease in CH50 levels (Caren & Rosenberg, 1966). In addition, it should be emphasized that hemolytic activity can change independent of circulating levels of complement proteins. There is normally an abundance of complement proteins in circulation and decreases in complement proteins do not become rate-limiting on hemolytic activity until there is more than a 50% decrease in complement protein levels (Atkinson & Frank, 1973). The decrease we observed in C4 levels may have been due in part to increased utilization (i.e., consumption due to the increased hemolytic activity). Lower C4 titers could also reflect decreased synthesis or increased catabolism induced by the high levels of adrenal corticoids in circulation.

The increase in hemolytic complement activity may be related to its capacity to respond as an acute phase reactant following tissue damage or in the early stages of infection. For example, hemolytic complement has been reported to increase in association with elevated erythrocyte sedimentation rate and C-reactive protein during rheumatic fever (Kaltreider, Caldwell & Ayoub, 1984). Hans Selye in his

early writings noted the many parallels between the stress response and the body's acute response to infection (Selye, 1950). The effect of stress may be particularly pronounced in the squirrel monkey because of its unusually high level of hemolytic activity and relatively active alternative complement or properdin pathway (Rommel, Bendure & Kalter, 1980; Rosenberg, Coe & Levine, 1982). Many factors can activate the alternative complement pathway, including IgA, bacterial endotoxin, and a variety of hematological substances. Resolution of whether the increased hemolytic activity was due to heightened activation of the classical or alternative pathway will require further study of the specific hemolytic activity of individual complement proteins. In addition, the clinical significance of the increased hemolytic activity will have to be determined. Increases in hemolytic complement activity have been observed in pregnant women and attributed to the high cortisol levels in the third trimester (Kitzmiller, Stoneburger, Yelenosky & Lucas, 1973). Pregnant squirrel monkeys maintain extemely high cortisol levels and often develop hemolytic anemia during late pregnancy (Coe et al., 1986; Rosenblum, 1978). In addition, following periods of stress, the squirrel monkey is prone to glomerulonephritis (Stills & Bullock, 1981), another disease that is associated with the complement system (Colten, 1980).

The decreased antibody responses in separated infant monkeys probably reflects two additional processes. In prior work we have observed up to a 23% decrease in circulating levels of IgG following separation of squirrel monkeys in the Alone condition (Coe, Cassayre, Rosenberg & Levine, 1988). Thus, there may be a general decrease in the synthesis of antibody, or an accelerated catabolism of antibody in the separated infant, as described in humans following pharmacological treatment with corticoids (Butler & Rossen, 1973). Endocrine activation could also interfere with the induction of the antibody response by affecting antigen-processing cells or the interaction between monocytes and lymphocytes (Munck et al., 1984). The latter effect would be in keeping with the preliminary observation that reunion with the mother on the day of antigen challenge did not restore the antibody response to normal in the previously-separated infants. That is, it was not necessary for the stressor to persist throughout the period of antibody production, only that the infant be in an altered state on the day of challenge. Although based on only 3 subjects in this experiment, we have observed a similar effect on 4 other infants evaluated in a Separation-Reunion condition (Coe, Rosenberg, Fischer & Levine, 1987).

The immunological changes in the squirrel monkey were greater than those observed in the rhesus macaque. Separated rhesus infants showed a decreased antibody response to ØX174, but the effect was less consistent. Rhesus infants also do not show decreases in circulating levels of IgG and complement proteins following maternal separation. The species difference may be due in part to the lower level of adrenal activation shown by the rhesus monkey. In the current experiment, the adrenal response of the separated rhesus infant did not even last 24 hours because it was left behind in a familiar cage after removal of the mother. Although we might have observed a somewhat greater endocrine and immune response in a novel separation environment, we believe that the species comparison illustrates the important theoretical point that large alterations in immune responses are more likely to occur in conjuction with sustained endocrine activation. Measurement of plasma cortisol levels provides one index of this endocrine activation, but it should be emphasized that even in the squirrel monkey, the amount of immune change did not directly correlate with cortisol titers in individual subjects.

Psychological manipulations and social disturbance appear to be particularly potent elicitors of immunological change in infants. Daily handling of rat pups has been shown to alter antibody responses later in life (Raymond, Reyes, Taokuda & Jones, 1986; Solomon, Levine & Kraft, 1968). Maternal separation has lingering effects on immune responses of many species when imposed as an early weaning condition (*mice*: Michaut, Dechambre, Doumerc, Lesourd, Deuillechabrolle & Moulias, 1981; *rats*: Ackerman, Keller, Schleifer, Shindledecker, Camerino, Hofer, Weiner & Stein, in press; *farm animals*: Kelley, 1980; *monkeys*: Laudenslager & Reite, 1985; Lubach, Coe, Ershler & Klopp, 1987). These data add to our knowledge on the extensive interactions among the nervous, endocrine, and immune systems, and suggest that infancy may be a vulnerable period in the life span when more persistent immunological changes can occur following psychological disturbance.

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